

Device: Shillehtek MPU6050 IMU

Data sheets:

- <https://howtomechatronics.com/tutorials/arduino/arduino-and-mpu6050-accelerometer-and-gyroscope-tutorial/>
- <https://invensense.tdk.com/wp-content/uploads/2015/02/MPU-6000-Register-Map1.pdf>
- Product specification:
<https://invensense.tdk.com/wp-content/uploads/2015/02/MPU-6000-Datasheet1.pdf>

Only using the accelerometer function. The accelerometer is meant to be used to sense vibrations and motion, so will be placed on a stationary object like a door or window. There should be no change in the orientation of the accelerometer.

Power consumption: Low power accelerometer mode current: 10 μ A at 1.25Hz, 20 μ A at 5Hz, 60 μ A at 20Hz, 110 μ A at 40Hz

Functions needed:

- Initialization - set up all registers and then get basis
- Filter may be needed - if accelerometer is noisy, cannot use interrupt anymore
 - Noise will cause false negatives
 - This accelerometer can implement an on module digital low pass filter
- No interrupt pin for accelerometer event
 - Interrupts only for FIFO overflow, I2C master, and data ready

Register Map:

Addr (Hex)	Addr (Dec.)	Register Name	Serial I/F	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0D	13	SELF_TEST_X	R/W	XA_TEST[4:2]			XG_TEST[4:0]				
0E	14	SELF_TEST_Y	R/W	YA_TEST[4:2]			YG_TEST[4:0]				
0F	15	SELF_TEST_Z	R/W	ZA_TEST[4:2]			ZG_TEST[4:0]				
10	16	SELF_TEST_A	R/W	RESERVED		XA_TEST[1:0]		YA_TEST[1:0]		ZA_TEST[1:0]	
19	25	SMPLRT_DIV	R/W	SMPLRT_DIV[7:0]							
1A	26	CONFIG	R/W	-	-	EXT_SYNC_SET[2:0]			DLPF_CFG[2:0]		
1B	27	GYRO_CONFIG	R/W	-	-	-	FS_SEL [1:0]		-	-	-
1C	28	ACCEL_CONFIG	R/W	XA_ST	YA_ST	ZA_ST	AFS_SEL[1:0]				
23	35	FIFO_EN	R/W	TEMP_FIFO_EN	XG_FIFO_EN	YG_FIFO_EN	ZG_FIFO_EN	ACCEL_FIFO_EN	SLV2_FIFO_EN	SLV1_FIFO_EN	SLV0_FIFO_EN
24	36	I2C_MST_CTRL	R/W	MULT_MST_EN	WAIT_FOR_ES	SLV_3_FIFO_EN	I2C_MST_P_NSR	I2C_MST_CLK[3:0]			
25	37	I2C_SLV0_ADDR	R/W	I2C_SLV0_RW	I2C_SLV0_ADDR[6:0]						
26	38	I2C_SLV0_REG	R/W	I2C_SLV0_REG[7:0]							
27	39	I2C_SLV0_CTRL	R/W	I2C_SLV0_EN	I2C_SLV0_BYTE_SW	I2C_SLV0_REG_DIS	I2C_SLV0_GRP	I2C_SLV0_LEN[3:0]			
28	40	I2C_SLV1_ADDR	R/W	I2C_SLV1_RW	I2C_SLV1_ADDR[6:0]						
29	41	I2C_SLV1_REG	R/W	I2C_SLV1_REG[7:0]							
2A	42	I2C_SLV1_CTRL	R/W	I2C_SLV1_EN	I2C_SLV1_BYTE_SW	I2C_SLV1_REG_DIS	I2C_SLV1_GRP	I2C_SLV1_LEN[3:0]			
2B	43	I2C_SLV2_ADDR	R/W	I2C_SLV2_RW	I2C_SLV2_ADDR[6:0]						
2C	44	I2C_SLV2_REG	R/W	I2C_SLV2_REG[7:0]							
2D	45	I2C_SLV2_CTRL	R/W	I2C_SLV2_EN	I2C_SLV2_BYTE_SW	I2C_SLV2_REG_DIS	I2C_SLV2_GRP	I2C_SLV2_LEN[3:0]			
2E	46	I2C_SLV3_ADDR	R/W	I2C_SLV3_RW	I2C_SLV3_ADDR[6:0]						
2F	47	I2C_SLV3_REG	R/W	I2C_SLV3_REG[7:0]							
30	48	I2C_SLV3_CTRL	R/W	I2C_SLV3_EN	I2C_SLV3_BYTE_SW	I2C_SLV3_REG_DIS	I2C_SLV3_GRP	I2C_SLV3_LEN[3:0]			
31	49	I2C_SLV4_ADDR	R/W	I2C_SLV4_RW	I2C_SLV4_ADDR[6:0]						
32	50	I2C_SLV4_REG	R/W	I2C_SLV4_REG[7:0]							
33	51	I2C_SLV4_DO	R/W	I2C_SLV4_DO[7:0]							
34	52	I2C_SLV4_CTRL	R/W	I2C_SLV4_EN	I2C_SLV4_INT_EN	I2C_SLV4_REG_DIS	I2C_MST_DLY[4:0]				
35	53	I2C_SLV4_DI	R	I2C_SLV4_DI[7:0]							
36	54	I2C_MST_STATUS	R	PASS_THROUGH	I2C_SLV4_DONE	I2C_LOST_ARB	I2C_SLV4_NACK	I2C_SLV3_NACK	I2C_SLV2_NACK	I2C_SLV1_NACK	I2C_SLV0_NACK
37	55	INT_PIN_CFG	R/W	INT_LEVEL	INT_OPEN	LATCH_INT_EN	INT_RD_CLEAR	FSYNC_INT_LEVEL	FSYNC_INT_EN	I2C_BYPASS_EN	-
38	56	INT_ENABLE	R/W	-	-	-	FIFO_OFLOW_EN	I2C_MST_INT_EN	-	-	DATA_RDY_EN
3A	58	INT_STATUS	R	-	-	-	FIFO_OFLOW_INT	I2C_MST_INT	-	-	DATA_RDY_INT

Addr (Hex)	Addr (Dec.)	Register Name	Serial I/F	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
3B	59	ACCEL_XOUT_H	R	ACCEL_XOUT[15:8]							
3C	60	ACCEL_XOUT_L	R	ACCEL_XOUT[7:0]							
3D	61	ACCEL_YOUT_H	R	ACCEL_YOUT[15:8]							
3E	62	ACCEL_YOUT_L	R	ACCEL_YOUT[7:0]							
3F	63	ACCEL_ZOUT_H	R	ACCEL_ZOUT[15:8]							
40	64	ACCEL_ZOUT_L	R	ACCEL_ZOUT[7:0]							
41	65	TEMP_OUT_H	R	TEMP_OUT[15:8]							
42	66	TEMP_OUT_L	R	TEMP_OUT[7:0]							
43	67	GYRO_XOUT_H	R	GYRO_XOUT[15:8]							
44	68	GYRO_XOUT_L	R	GYRO_XOUT[7:0]							
45	69	GYRO_YOUT_H	R	GYRO_YOUT[15:8]							
46	70	GYRO_YOUT_L	R	GYRO_YOUT[7:0]							
47	71	GYRO_ZOUT_H	R	GYRO_ZOUT[15:8]							
48	72	GYRO_ZOUT_L	R	GYRO_ZOUT[7:0]							
49	73	EXT_SENS_DATA_00	R	EXT_SENS_DATA_00[7:0]							
4A	74	EXT_SENS_DATA_01	R	EXT_SENS_DATA_01[7:0]							
4B	75	EXT_SENS_DATA_02	R	EXT_SENS_DATA_02[7:0]							
4C	76	EXT_SENS_DATA_03	R	EXT_SENS_DATA_03[7:0]							
4D	77	EXT_SENS_DATA_04	R	EXT_SENS_DATA_04[7:0]							
4E	78	EXT_SENS_DATA_05	R	EXT_SENS_DATA_05[7:0]							
4F	79	EXT_SENS_DATA_06	R	EXT_SENS_DATA_06[7:0]							
50	80	EXT_SENS_DATA_07	R	EXT_SENS_DATA_07[7:0]							
51	81	EXT_SENS_DATA_08	R	EXT_SENS_DATA_08[7:0]							
52	82	EXT_SENS_DATA_09	R	EXT_SENS_DATA_09[7:0]							
53	83	EXT_SENS_DATA_10	R	EXT_SENS_DATA_10[7:0]							
54	84	EXT_SENS_DATA_11	R	EXT_SENS_DATA_11[7:0]							
55	85	EXT_SENS_DATA_12	R	EXT_SENS_DATA_12[7:0]							
56	86	EXT_SENS_DATA_13	R	EXT_SENS_DATA_13[7:0]							
57	87	EXT_SENS_DATA_14	R	EXT_SENS_DATA_14[7:0]							
58	88	EXT_SENS_DATA_15	R	EXT_SENS_DATA_15[7:0]							
59	89	EXT_SENS_DATA_16	R	EXT_SENS_DATA_16[7:0]							
5A	90	EXT_SENS_DATA_17	R	EXT_SENS_DATA_17[7:0]							
5B	91	EXT_SENS_DATA_18	R	EXT_SENS_DATA_18[7:0]							
5C	92	EXT_SENS_DATA_19	R	EXT_SENS_DATA_19[7:0]							
5D	93	EXT_SENS_DATA_20	R	EXT_SENS_DATA_20[7:0]							
5E	94	EXT_SENS_DATA_21	R	EXT_SENS_DATA_21[7:0]							
5F	95	EXT_SENS_DATA_22	R	EXT_SENS_DATA_22[7:0]							
60	96	EXT_SENS_DATA_23	R	EXT_SENS_DATA_23[7:0]							
63	99	I2C_SLV0_DO	R/W	I2C_SLV0_DO[7:0]							
64	100	I2C_SLV1_DO	R/W	I2C_SLV1_DO[7:0]							
65	101	I2C_SLV2_DO	R/W	I2C_SLV2_DO[7:0]							
66	102	I2C_SLV3_DO	R/W	I2C_SLV3_DO[7:0]							

Addr (Hex)	Addr (Dec.)	Register Name	Serial I/F	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
67	103	I2C_MST_DELAY_CTRL	R/W	DELAY_ES_SHADOW	-	-	I2C_SLV4_DLY_EN	I2C_SLV3_DLY_EN	I2C_SLV2_DLY_EN	I2C_SLV1_DLY_EN	I2C_SLV0_DLY_EN
68	104	SIGNAL_PATH_RESET	R/W	-	-	-	-	-	GYRO_RESET	ACCEL_RESET	TEMP_RESET
6A	106	USER_CTRL	R/W	-	FIFO_EN	I2C_MST_EN	I2C_IF_DIS	-	FIFO_RESET	I2C_MST_RESET	SIG_COND_RESET
6B	107	PWR_MGMT_1	R/W	DEVICE_RESET	SLEEP	CYCLE	-	TEMP_DIS	CLKSEL[2:0]		
6C	108	PWR_MGMT_2	R/W	LP_WAKE_CTRL[1:0]		STBY_XA	STBY_YA	STBY_ZA	STBY_XG	STBY_YG	STBY_ZG
72	114	FIFO_COUNTH	R/W	FIFO_COUNT[15:8]							
73	115	FIFO_COUNTL	R/W	FIFO_COUNT[7:0]							
74	116	FIFO_R_W	R/W	FIFO_DATA[7:0]							
75	117	WHO_AM_I	R	-	WHO_AM_I[6:1]						-

Configuration Settings:

- 0x1A - Config
- 0x1B - Gyro Config
 - Not using Gyro
- 0x1C -Accel Config
- 0x23 - FIFO Enable
 - FIFO buffer is unneeded - using accelerometer as an interrupt
- 0x24 - I2C Master Control
- 0x37 - Interrupt Pin Config
- 0x38 - Interrupt Enable
- 0x6A - User Control
- 0x6B - Power Management 1
- 0x6C - Power Management 2

I2C

- 0x25 - I2C Slave0 Address
- 0x27 - I2C Slave0 Control
- 0x33 - I2C Master Status

Configuration

0x1A - Config

Register (Hex)	Register (Decimal)	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1A	26	-	-	EXT_SYNC_SET[2:0]			DLPF_CFG[2:0]		

Frame synchronization and digital low pass filter.

Accel data can be quite noisy (especially if the power supply is noisy), so digital low pass filter will help with that.

Not going to be using frame synchronization.

DLPF_CFG	Accelerometer ($F_s = 1\text{kHz}$)	
	Bandwidth (Hz)	Delay (ms)
0	260	0
1	184	2.0
2	94	3.0
3	44	4.9
4	21	8.5
5	10	13.8
6	5	19.0
7	RESERVED	

Only using the accelerometer as an interrupt, so a higher bandwidth is acceptable. However, do not want high latency between an event and when video actually is recorded. Will need to find what is acceptable through testing.

0x1C - Accelerometer Config

Register (Hex)	Register (Decimal)	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1C	28	XA_ST	YA_ST	ZA_ST	AFS_SEL[1:0]		-		

Self testing - bits 7-5

AFS_SEL - full scale range

- $\pm 2g$ is probably sufficient - need to test to make sure

0x24 - I2C Master Control

Register (Hex)	Register (Decimal)	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
24	36	MULT_MST_EN	WAIT_FOR_ES	SLV_3_FIFO_EN	I2C_MST_P_NSR	I2C_MST_CLK[3:0]			

Only 1 master - MULT_MST_EN = 0

I2C_MST_CLK -

I2C_MST_CLK	I ² C Master Clock Speed	8MHz Clock Divider
0	348 kHz	23
1	333 kHz	24
2	320 kHz	25
3	308 kHz	26
4	296 kHz	27
5	286 kHz	28
6	276 kHz	29
7	267 kHz	30
8	258 kHz	31
9	500 kHz	16
10	471 kHz	17
11	444 kHz	18
12	421 kHz	19
13	400 kHz	20
14	381 kHz	21
15	364 kHz	22

- Specific speed will be determined by what STM32 is set up for

0x37 - Interrupt Pin Config

Register (Hex)	Register (Decimal)	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
37	55	INT_LEVEL	INT_OPEN	LATCH_INT_EN	INT_RD_CLEAR	FSYNC_INT_LEVEL	FSYNC_INT_EN	I2C_BYPASS_EN	-

Register set: 0x00

- INT_LEVEL: 0 - active high
- INT_OPEN: 0 - push-pull
 - Push-pull or open drain
 - Not connecting multiple devices on same line or voltage translation (high signal is equal to the voltage provided by the pull up resistor), plus just breadboarding so would prefer not having a pull up resistor
- LATCH_INT_EN: 0 - pulse
 - 0 - 50us pulse
 - 1 - INT pin held high until interrupt is cleared
 - Right now do pulse
- INT_RD_CLEAR: 0
 - 0: Interrupt status bits only cleared by reading INT_STATUS
 - 1: Interrupt status bit cleared on any read
- Not using FSYNC
- I2C_BYPASS_EN: 0

- 0: host application processor not able to directly access auxiliary I2C bus
- 1: host application processor able to directly access auxiliary I2C bus on MPU-60X0

0x38 - Interrupt Enable

Register (Hex)	Register (Decimal)	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
38	56		-		FIFO_OFLOW_EN	I2C_MST_INT_EN	-	-	DATA_RDY_EN

0x00

- FIFO_OFLOW_EN: 0 - not using FIFO
- I2C_MST_INT_EN: 0 - I2C master interrupt sources cannot generate an interrupt
- DATA_RDY_EN: 0 - do not want an in interrupt when data is ready

0x6A - User Control

Register (Hex)	Register (Decimal)	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
6A	106	-	FIFO_EN	I2C_MST_EN	I2C_IF_DIS	-	FIFO_RESET	I2C_MST_RESET	SIG_COND_RESET

0x00

- FIFO_EN: 0 - not using FIFO
- I2C_MST_EN: 0 - not using MPU-60X0 as a master
- I2C_IF_DIS: 0

0x6B - Power Management 1

Register (Hex)	Register (Decimal)	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
6B	107	DEVICE_RESET	SLEEP	CYCLE	-	TEMP_DIS	CLKSEL[2:0]		

- SLEEP:
 - 1 is low power mode
- CYCLE:
 - 1: SLEEP disabled, in Cycle Mode. Device cycles b/w sleeping and waking up to take a data sample at rate defined by LP_WAKE_CTRL (reg 0x6C)
- TEMP_DIS: 1 - disable temperature sensor

- CLKSEL: 0 - internal 8MHz oscillator

CLKSEL	Clock Source
0	Internal 8MHz oscillator
1	PLL with X axis gyroscope reference
2	PLL with Y axis gyroscope reference
3	PLL with Z axis gyroscope reference
4	PLL with external 32.768kHz reference
5	PLL with external 19.2MHz reference
6	Reserved
7	Stops the clock and keeps the timing generator in reset

0X6C - Power Management 2

Register (Hex)	Register (Decimal)	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
6C	108	LP_WAKE_CTRL[1:0]		STBY_XA	STBY_YA	STBY_ZA	STBY_XG	STBY_YG	STBY_ZG

- LP_WAKE_CTRL:
 - Rate of wake up to take data sample
- Bits 2:0 - standby for gyroscope axis

The MPU-60X0 can be put into Accelerometer Only Low Power Mode using the following steps

 - Set CYCLE bit to 1
 - Set SLEEP bit to 0
 - Set TEMP_DIS bit to 1
 - Set STBY_XG, STBY_YG, STBY_ZG bits to 1
- All of the above bits can be found in Power Management 1 register (Register 107).

Operating Procedure

Initialization

- 1) Turn on accelerometer
- 2) Read from WHO_AM_I register - verify that it is correct
 - a) If not stop
- 3) Write all configuration registers during initialization
- 4) Read back configuration registers to ensure written to correctly

FIFO Buffer Management

- FIFO buffer - can only read out of it 8 bits at a time
- 1024-byte FIFO
- Going to have a fifo buffer interrupt - FIFO_OFLOW_INT

- Timed reads (timer interrupt). FIFO is 1024 bytes
 - Need to calculate time to fill
 - Accelerometer data - 3 axis, each axis is 16 bits (2 bytes). 6 bytes per sample
 - $1024/(6 \times \text{sample_rate})$ to get time till FIFO is full
- 2 registers - FIFO count tell current count of data in FIFO